

Global Grouped Checkpoint Optimization

Short technical report | September 2026

Executive summary

We built and validated a global checkpoint policy optimizer for multi-job training workloads. Instead of tuning each job independently, the optimizer partitions jobs into groups and jointly selects checkpoint route, frequency, and phase under shared network, storage, donor, durability, and recovery constraints. The simulator integration enforces group-scoped donor membership and provides runtime evidence for the policy decisions.

What we implemented

The optimizer searches a finite catalogue of direct and peer modes, evaluates feasible partitions, and exports an auditable contract containing `group_id`, `route`, `allowed donors`, `interval_s`, and `phase_s`. The simulator now supports group-aware donor filtering through `checkpointing/group_policy.py` and the cross-job registry. Runtime budget checks, recovery audits, boundary checks, and policy bridge tests were added or exercised. The final test status was 225 simulator tests passing, with the optimizer test suite also passing.

Experimental design

We compared three policies on the three-job cross-job smoke workload using seeds 7, 8, and 9: (A) direct checkpoints with no grouping, (B) one group with peer enabled and phase 0, and (C) one group with peer enabled and a 15-second phase for jobB. We measured simulator time, peer rate, fallback count, own-flush refusals, donor failures, and flush duration.

Policy	Mean time (s)	Peer rate	Fallback	Own flush	Donor failed
A: no grouping	212.34	0.000	7.33	0.00	0.00
B: grouped, phase 0	220.36	0.139	6.33	16.00	2.00
C: grouped + phased	226.73	0.194	6.00	10.67	4.67

Table 1. Mean values over seeds 7, 8, and 9.

Interpretation

The results are internally consistent with a shared-resource system. Grouping and phasing increased peer utilization from 0.000 to 0.194 and reduced fallback from 7.33 to 6.00. Phasing reduced own-flush refusals relative to the unphased grouped policy. At the same time, contention and donor exposure increased, so mean simulation time rose from 212.34 seconds to 226.73 seconds. The current evidence therefore supports a multi-objective interpretation: peer utilization and fallback can improve while makespan worsens.

Main contribution

The project delivers a global, group-aware checkpoint policy pipeline that connects analytical search to event-driven simulation. It makes the system trade-off explicit and measurable instead of optimizing a single job in isolation. The policy contract provides a clean interface between the optimizer and runtime implementation, while the simulator checks donor membership, shared budgets, checkpoint waves, and recovery accounting.

Limitations and next step

The runtime penalty model was calibrated from a small three-job smoke workload, so it should not be treated as a production predictor for hundreds of jobs. The peer-required mode currently uses a selection bias to force peer usage; a clean final implementation should express this as a native MILP constraint and report the unmodified analytical objective separately. The next experiment should scale group sizes to 2, 3, 5, and 10 jobs, run more seeds, and compare Pareto trade-offs under realistic failure and bandwidth scenarios.

Presentation claim

A defensible claim for a group presentation is: “The proposed optimizer performs global, constraint-aware selection of grouping, checkpoint frequency, phase, and peer route. In the validated smoke workload it increases peer utilization and reduces fallback, while exposing a measurable contention trade-off in makespan and donor failures.”